AutoVox User Manual

Revision 1 - April 6, 2026

Team Five of a Kind

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Introduction & Product Description

Welcome to AutoVox! Within this manual, you will find detailed operational instructions about the use of AutoVox, ranging from basic use to advanced tips. Installation, Terminology, Use Cases, and FAQs are covered in turn, but it is not necessary to read in order; feel free to skip sections to find needed information.

About AutoVox

AutoVox is a lightweight tool to assist in the creation of designs for voxel-based, multi-material 3D printing. AutoVox aims to speed up development by providing an interface allowing mass edits to many voxels at a time, and automating conversion from an STL file to a set of voxels.

About this Manual

Where to find information:

- ➔ *Installation:* Covers installation instructions.
- ➔ *Terminology:* Explains terms specific to AutoVox used throughout this manual.
- ➔ *Use Cases & How-To:* Detailed instructions about how to complete different tasks. Sub-headers indicate tasks discussed within.
- ➔ *FAQs:* Typical sources of confusion and customization discussed in a question-and-answer format. Consult for quick answers to common questions.

This manual uses two signifiers to highlight information to readers when skimming, which are explained below.

Notice: this icon is used to draw attention to important facts that may cause issues if users are unaware.

Tip: this icon signifies advanced information that may be helpful to more familiar users.

Other important information may be emphasized by **bold** or *italics*.

Installation

The app offers a standalone version, built and tested on MacOS. If you require a windows install, and the current version does not work you will need to locally clone the repo and follow these steps.

Prerequisites:

Install Node.js 18 or newer. For the packaged app, install Python 3.9+ (3.10+ matches the rest of the project). On first launch the app uses that interpreter to create a private virtual environment and install the bundled FastAPI backend from requirements.txt. On Windows, either enable Add python.exe to PATH during setup or keep the full path to python.exe handy for the setup screen if the app asks for it.

To build, run the following:

```
cd src/app

npm install

npm run pack
```

This builds the Vite frontend with VITE_API_BASE_URL=http://127.0.0.1:8765, then runs electron-builder. Outputs go to src/app/release/ (for Windows targets typically an NSIS.exe and a zip). Building a Windows installer is intended to be done on Windows (or in CI on a Windows runner); the macOS build configuration in this repo targets dmg and zip, not cross-compiled Windows artifacts.

First run: Expect a short setup phase the first time: Python check, venv (virtual python environment) creation, and pip install into app user data. Later launches reuse this environment.

Develop without packaging. Start the backend as in the main README (uvicorn in src/backend on port 8000). Then from src/app, after npm install, run npm run dev so Electron loads the Vite dev server and talks to the API on localhost.

Terminology

This section covers the terms used throughout this manual.

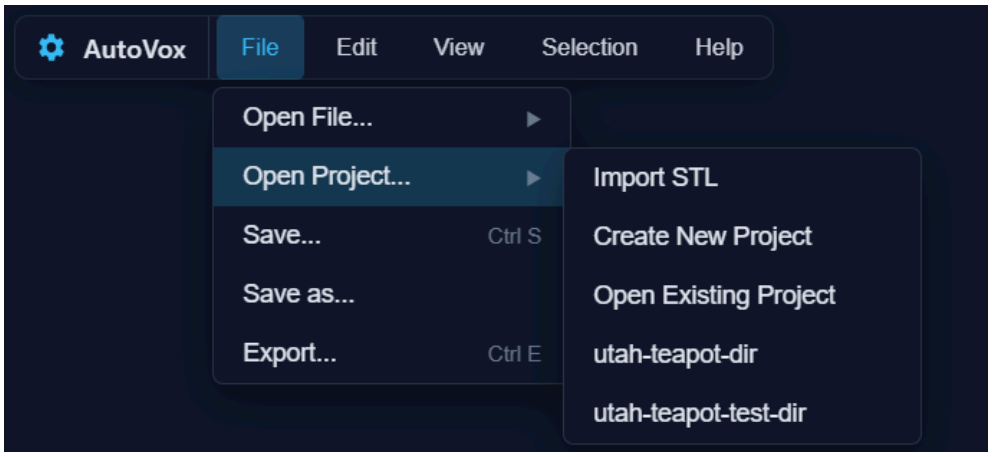
- *Project / 3D Voxel Model:* These terms are used interchangeably to refer to the set of voxels created after an STL file is imported.
- *Model View:* Refers to the rotatable, 3D voxel model shown in the centre of the UI.
- *Layer:* A set of voxels that share the same Z-coordinate. It is these segments that can be navigated between to edit groups of voxels within.
- *Layer Editor:* The right pop-out interface where changes to the model are made, within a selected layer.
- *Layer View:* The zoomable 2D panel located at the top of the layer editor interface where voxels are selected and their status is visually shown.
- *Partition:* A 3D chunk of the full model that is edited separately from others. A model may be split into partitions based on how many voxels are created to allow for easier editing management.

Use Cases & How-To

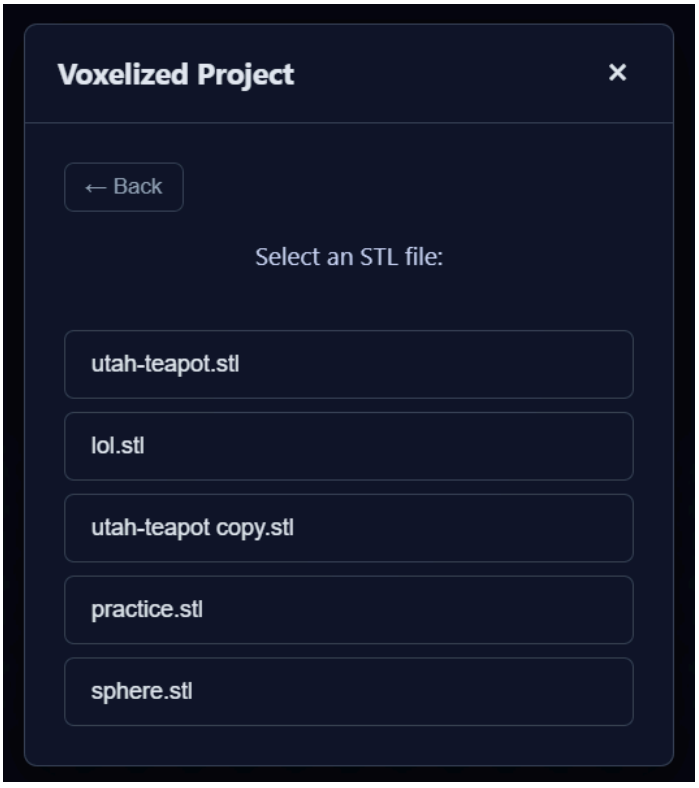
This section contains simple step-by-step instructions regarding the primary use case and workflow of AutoVox, from file import to editing and export.

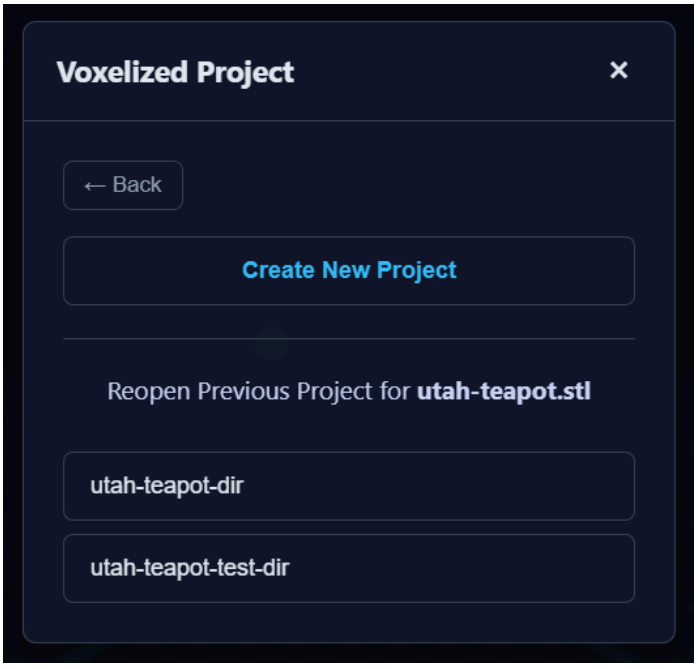
Project Creation

To create a new project, begin by importing an STL file via the 'File' drop-down menu (File > Open Project… > Import STL).

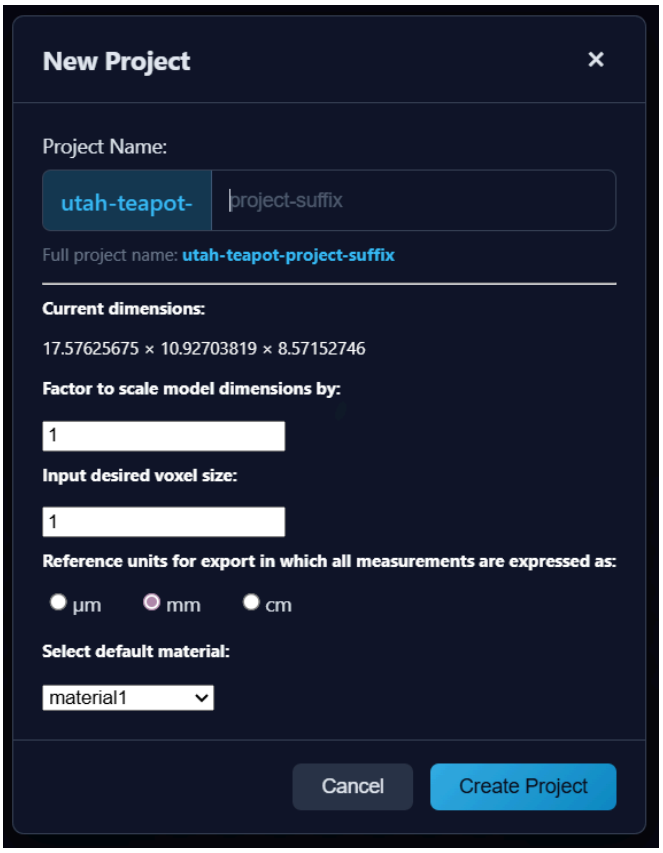


After the STL file is rendered within AutoVox, create a new project via (File > Open Project… > Create New Project). This will result in the following creation menus being displayed.





The first menu allows selection of available STL files (e.g. STL files you have previously uploaded). Clicking on the STL file you uploaded to select it brings you to the second menu. If you have previously created voxel models with the STL file, you can reopen them from here, or create a new project. Clicking on "Create New Project" brings you to the final customization menu shown on the next page.



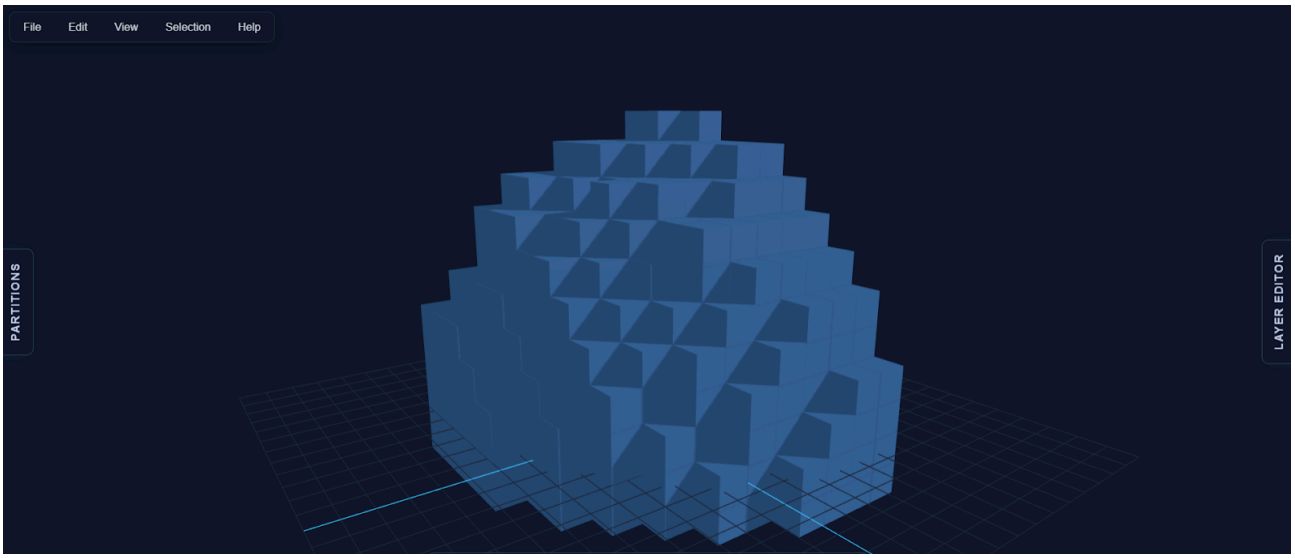
Within this menu, a project name and the units of the model and all measurements can be specified.

> Note that as STL files natively have no units, you must specify units based on your design, e.g. 1 STL unit = 1mm.

To control voxel size, specify the units of voxels, as well as their scale. The combination of STL file units and voxel units will determine how many voxels are created for a given project.

Each project must also have a defined default material for all voxels, which is entered upon project creation. Upon creation, all voxels within the project and any new voxels added will begin with this material.

When finished, click ‘Create Project’ at the bottom of the creation menu. AutoVox will then automatically load the new project, as shown below.



From this screen, orientation of the resulting 3D voxel model can be adjusted by holding left click and dragging naturally across the screen.

> Once a project is created, *STL units, voxel units/scale, and default voxel properties cannot be adjusted!* If you find you have made an error, you will be required to create a new project from your STL file.

Using the Layer Editor

All edits to the project occur within the layer editor. To open it, click on any voxel on the project. This will open the layer that voxel belongs to within the layer editor to the right of the screen.

1) Toggle buttons to switch between voxel selection styles. ‘Click’ allows selecting one voxel at a time, while ‘Lasso’ allows drawing shapes around voxels to be selected.

2) Button to toggle between voxel editing mode and add/remove voxel mode.

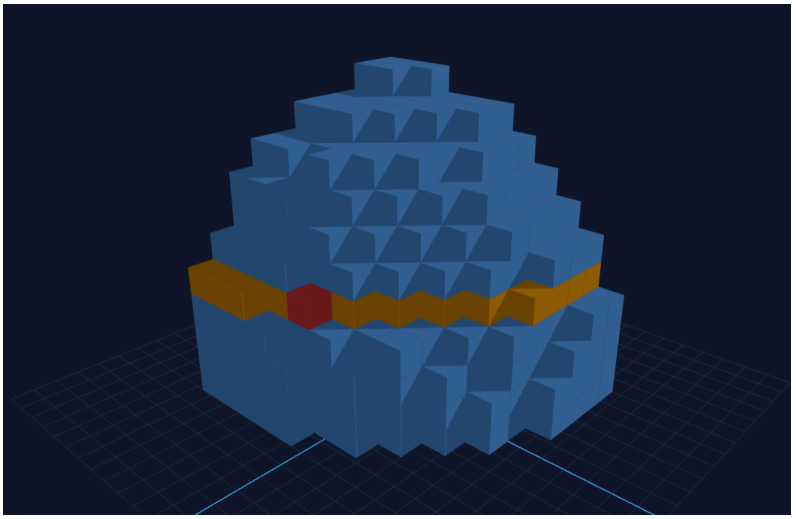
3) The layer view panel, showing a layer with multiple different materials (represented by squares with different colours) and magnetizations (checkmarks within each square).

4) Buttons to navigate up and down between layers.

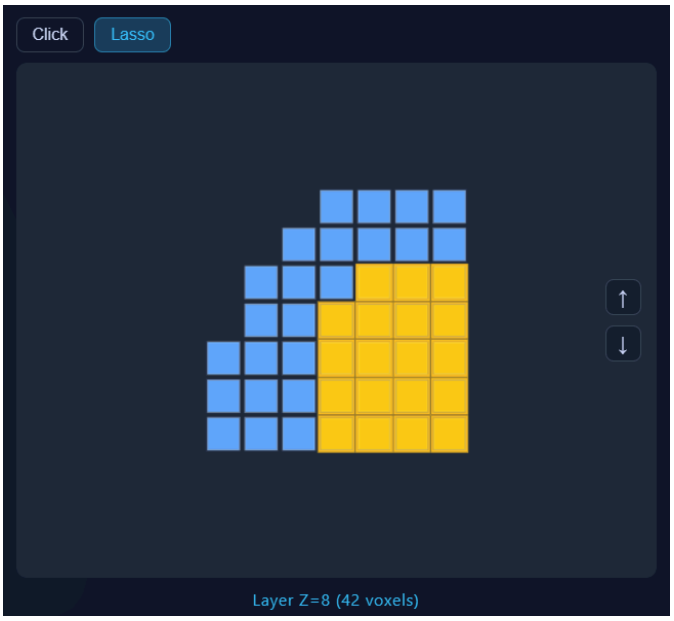
5) Indicator for currently selected voxel(s).

6) Buttons to confirm an update to a set of voxels.

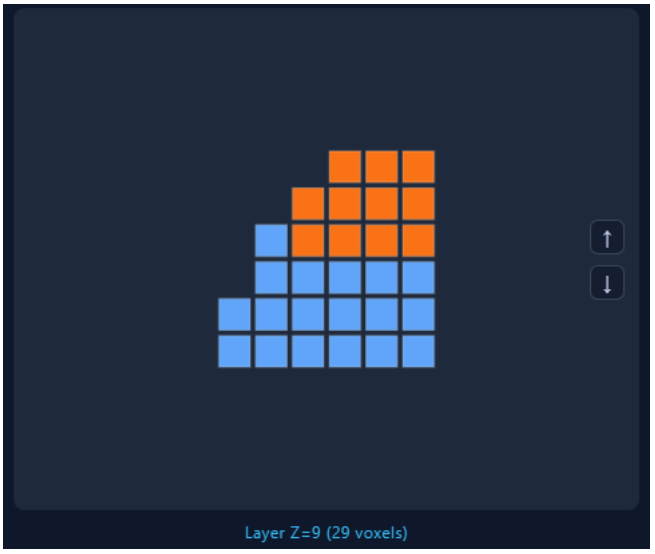
A view of the 3D Model Viewer after clicking on a voxel. The layer currently being viewed within the layer editor is highlighted.



➔ Selecting Voxels: single voxels can be selected one at a time by left clicking on them within the layer view. To select multiple voxels at a time, hold left click and draw around the voxels you wish to select. Selected voxels are then shown as yellow.

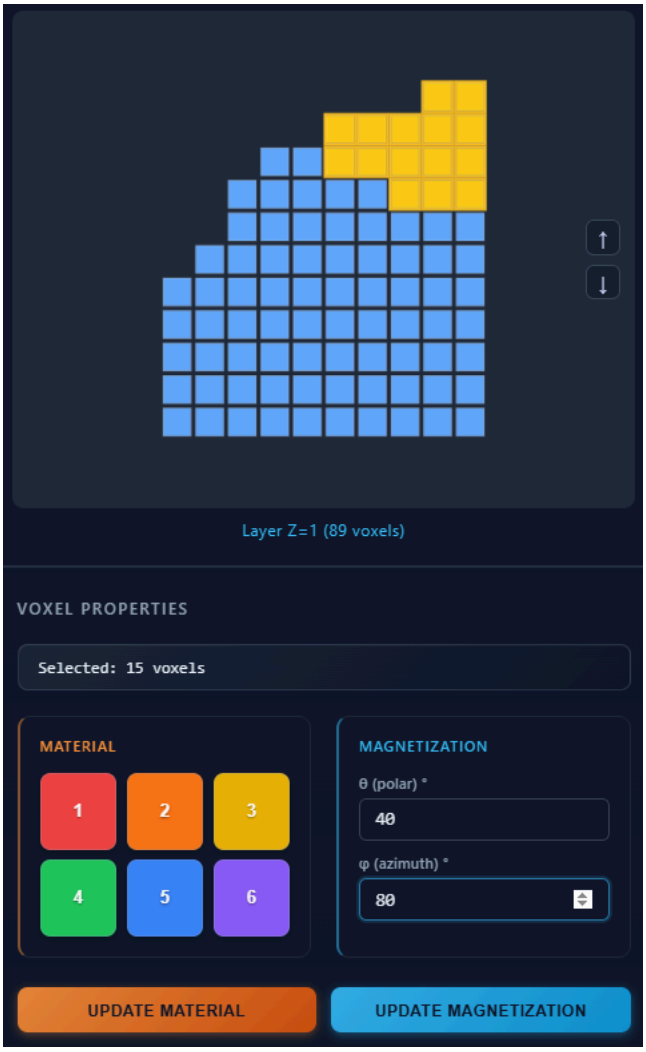


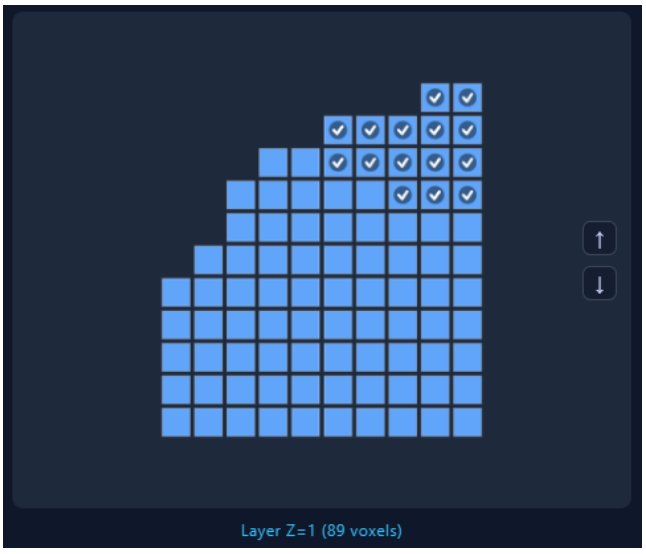
➔ Assigning Materials: After voxels are selected, their material can be edited by clicking on one of the material squares below the layer view under 'Materials', and then clicking confirm. Changes are immediately reflected within the layer view with the voxels changing color to reflect their new material.



Before and after material assignment.

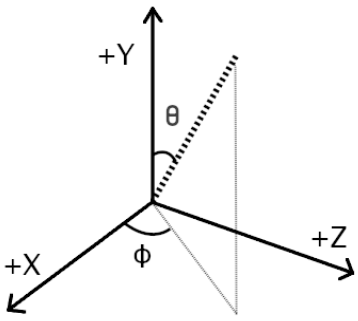
➔ <u>Assigning Magnetizations:</u> After voxels are selected, their magnetization can be edited by entering a polar and azimuthal angle in the two labelled text boxes, and then clicking confirm. Changes are immediately reflected within the layer view with the voxels gaining a checkmark to reflect their new state.



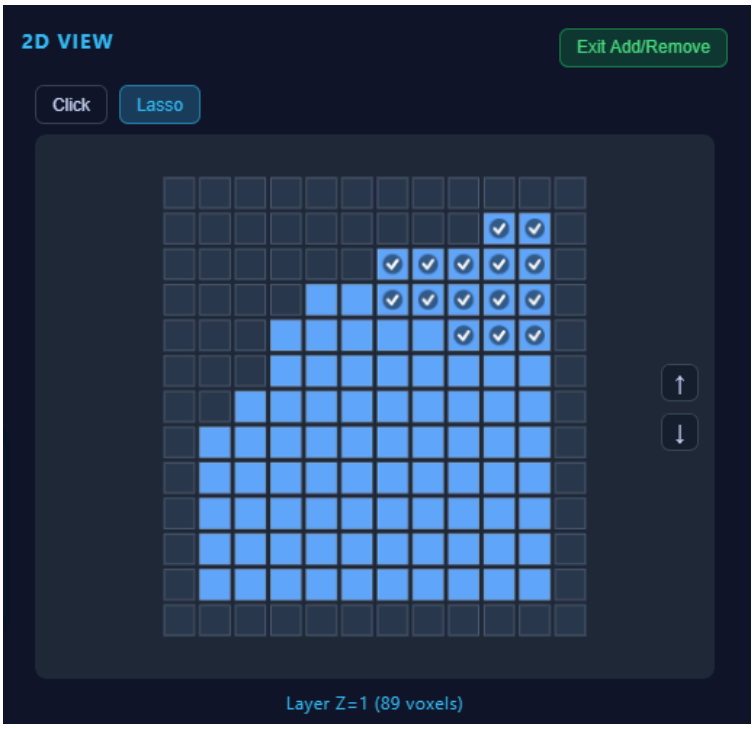


Before and after magnetization assignment.

Note that the polar angle and azimuthal angle fields are restricted to [0, 180] and [0, 360] respectively to ensure angle uniqueness. See the diagram below to see where angles are measured from for each voxel.



- ➔ <u>Adding and Deleting Voxels:</u> Voxels can be added and deleted by first enabling 'Add/Remove Voxels' mode by left clicking on the upper right button marked as such, and then right and left clicking respectively on grid squares within the layer view, or using lasso select with left or right click. Recall that voxels added will contain the default properties for the project configured at time of creation. The mode can be exited by left clicking on the button again.



Additional Tips:

- ➔ Zoom in/out within the layer view for more precise selection within large layers.
- ➔ Quickly switch layers either by directly clicking on the layer within the model viewer, or navigate up or down one layer via the ↑ and ↓ arrows on the right side of the layer view.
- ➔ If a mistake is made, undo via (Ctrl+Z) and redo via (Ctrl+Y).

Working with Partitions

If a project is sufficiently large, the project will be split into a grid of partitions that can be swapped between. Each effectively functions as its own model. Below are some important facts to know.

- ➔ Open the partition menu by clicking on the respective button to the left of the screen. Once open, name partitions by double-clicking on any partition in the menu.
- ➔ Switch between partitions by left-clicking on any of the listings within the menu.
- ➔ In the current release, AutoVox does not stop you from adding voxels that may interfere with adjacent partitions. **You must be aware of where partitions match up,** otherwise you may have a final export containing duplicate voxels.

> ⚠ **Important:** As partitions are saved within separate files, when switching between partitions, undo/redo history is LOST!

Project Export

Exporting projects is as simple as navigating to (File > Export). This will download a .csv with columns reflecting all voxel x, y, and z coordinates, their assigned material as an integer, followed by polar and azimuthal angles.

FAQs

- ➔ Can I alter what axis layers are created upon?
  - ◆ No. Within the current release, layers are only created upon the vertical axis (Y). However, this feature may make its way into a future release.
- ➔ Are materials per-project?
  - ◆ No. The same set of materials transfer across all projects.
- ➔ How do I navigate between partitions?
  - ◆ Click on the partition tab to the left of the screen and left click on any of the partitions listed. Remember that doing so will clear your undo/redo history on the partition you are currently working on!
- ➔ Can partitions be renamed?
  - ◆ Yes! Double click on partitions to rename them.
- ➔ How do I zoom in/out in the 3D model viewer?
  - ◆ Pinch two fingers together (zoom out) or move them apart (zoom in) anywhere on the screen. The same action works within the layer editor to zoom in or out.
- ➔ What should I do if I find a bug?
  - ◆ Create an issue on the GitHub repository for AutoVox here! Try your best to describe the unintended behavior, as well as how to recreate the bug if possible.